Technical Assignment (With Plivo API)

Position Overview

We are seeking a Forward Deployed Engineer (FDE) to join our team to build communication-forward applications using APIs like Plivo. This assignment evaluates your ability to integrate APIs, handle call logic, authenticate callers, and design a multi-level IVR.

Technical Assignment

Scenario

InspireWorks wants a demo IVR system that demonstrates Plivo's voice API capabilities. The system should:

- Make an outbound call from the given Plivo phone number to your phone number.
- Upon answer, authenticate the caller via OTP:
  - Bot prompts the caller to enter a 4-digit OTP.
  - The OTP is the caller's birthdate in DDMM format (hardcoded by you, e.g. March 15 → 1503).
  - If incorrect OTP is entered, the bot re-prompts until the correct OTP is received.
- Once authenticated, present a multi-level IVR menu:
  - Level 1: Language selection
    - Press 1 for English
    - Press 2 for Spanish
  - Level 2 (after language selection):
    - Press 1 to play a short audio message
    - Press 2 to connect to a live associate (use a placeholder number)

The goal is to demonstrate caller authentication, multi-level voice interaction, branching logic, and Plivo API integration.

Implementation Requirements

1. Outbound Call

- Build an endpoint or script to initiate an outbound call using Plivo's API.
- Target phone number can be provided via a simple UI or configuration variable.

2. OTP Authentication Layer

- On call answer, prompt the caller to enter a 4-digit OTP using DTMF input.
- The OTP should be your birthdate in DDMM format — hardcode this value in your application.
- If the caller enters the wrong OTP, the bot must re-prompt until the correct OTP is entered.

- Only after a correct OTP should the caller be allowed into the IVR menu.

3. IVR Menu

- Level 1: Prompt caller to select a language (English / Spanish).
- Level 2: Depending on language choice:
  - Press 1 → Play an audio file (e.g. MP3 hosted publicly).
  - Press 2 → Forward call to an associate (can be a placeholder number).
- Use Plivo XML for call flow.

4. Multi-level Flow Handling

- Correctly handle DTMF input across all levels (OTP, language, action).
- Implement branching logic based on user input.
- Handle invalid inputs gracefully (e.g. repeat the current prompt).

5. Optional Frontend

- Simple web page or CLI to trigger the outbound call.
- Not required, but helpful for demonstration.

Deliverables

Working Application

- Script / web app that triggers outbound calls and supports the full IVR flow.
- Demonstrate OTP authentication, Level 1 and Level 2 interactions.

Code Repository

- GitHub or zip file with source code.
- Include README with:
  - Setup instructions
  - Required Plivo credentials (Auth ID / Auth Token)
  - Steps to run and test

Short Demo Video (3–5 min)

- Show a call being made
- Demonstrate OTP prompt — enter a wrong OTP first, then the correct one
- Navigate Level 1 → Level 2 menus
- Demonstrate audio playback and call forwarding

Also attach the receiver's (your) phone number that you used in the mail

Notes / Tips

- For audio playback, use a publicly accessible MP3 file or Plivo's default audio.
- The live associate number can be a placeholder or your own test number.
- OTP is your own birthdate in DDMM format — hardcode it. No database needed.

Auth ID - MAMTAWMGI0MZCTNTYZZS
Auth Token - M2Y2MzllMWEtOGU5Ny00YzYxLWFkNzItZmE5ZmNl
Plivo number to be used - +918035454161
Plivo number to be used as the Live Associate number - 02264236412